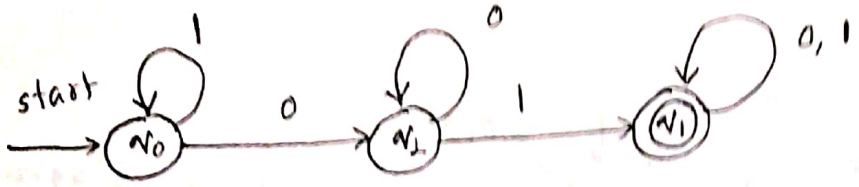
Q: Construct a DFA that accepts all and only the strings of 0's and 1's that have the sequence 01 somewhere in the string.

Ans:



Here, $Q = \{q_0, q_1, q_2\}$

$\Sigma = \{0, 1\}$

$\delta(q_0, 1) = q_0$, $\delta(q_0, 0) = q_2$

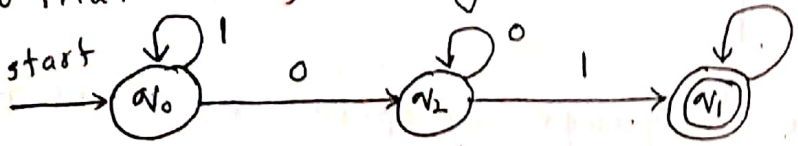
$\delta(q_2, 0) = q_2$, $\delta(q_2, 1) = q_1$

$\delta(q_1, 0) = q_1$, $\delta(q_1, 1) = q_1$

starting state, q_0

Final state, $F = \{q_1\}$

Ques: show that the following DFA accept string 10011



Ans:

$\delta(q_0, 1) = q_0$

$\delta(q_0, 10) = \delta(\delta(q_0, 1), 0) = \delta(q_0, 0) = q_2$

$\delta(q_0, 100) = \delta(\delta(q_0, 10), 0) = \delta(q_2, 0) = q_2$

$\delta(q_0, 1001) = \delta(\delta(q_0, 100), 1) = \delta(q_2, 1) = q_1$

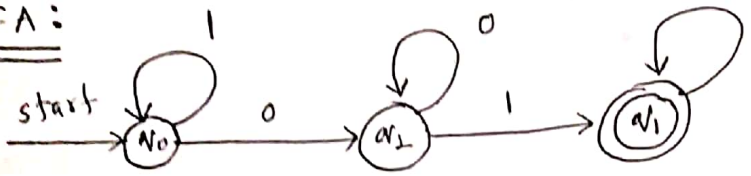
$\delta(q_0, 10011) = \delta(\delta(q_0, 1001), 1) = \delta(q_1, 1) = q_1$

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Transition Table:

A transition table which is a tabular listing of function.

DFA:



⇐ Transition diagram

	0	1
→ q_0	q_2	q_0
* q_1	q_1	q_1
q_2	q_2	q_1

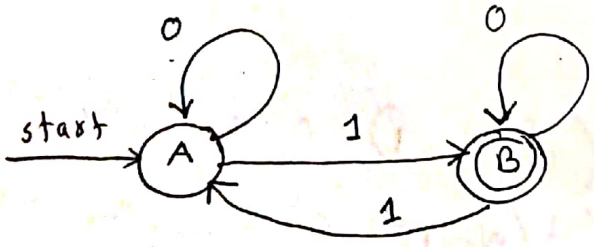
⇐ Transition Table

Ques:

	0	1
→ A	A	B
* B	B	A

Describe the language accepted by this DFA.

Ans:



This DFA supports all the strings, ~~that contains ending with 1~~ which have ~~at least one 1 at any position~~ odd number 1's

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